

Dear Editor,

Enclosed please find our manuscript "Is Stephen Curry really a guard? New perspective on players typology using functional data analysis". Using data from the NBA, we present a novel representation of NBA players' shooting patterns based on Functional Data Analysis. Each player's charts shots are treated as smooth functional data. Using a multivariate functional principal components analysis decomposition, we capture the primary modes of variability in shooting patterns and quantify individual deviations from the average behavior. Using clustering algorithms, we are able to construct a data-driven taxonomy of players. Our empirical clusters capture aspects of playing style not reflected in official designations. The methodology provides a flexible, interpretable and continuous framework for analyzing player tendencies, with potential applications in coaching, scouting and historical player or match comparisons.

We hope that it could be considered for a reviewing process for the *Journal of Quantitative Analysis in Sports*.

Sincerely,

Steven Golovkine and Edward Gunning